



Katholieke
Universiteit
Leuven

Department of
Electrical Engineering

PROJECT 2: ANALYZING A CRYPTOSYSTEM BASED ON THE FINITE FIELD ISOMORPHISM PROBLEM

Computer Algebra for Cryptography (B-KUL-H0E74A)

Gust De Wit (r0948039)

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1 Finite Fields and Irreducible Polynomials

1.1 Finding an isomorphism

Alternate formulation First, we will show that finding an isomorphism between $\mathbb{X}\langle x \rangle = \mathbb{F}_p[z]/f(z)$ and $\mathbb{Y}\langle y \rangle = \mathbb{F}_p[z]/g(z)$ is equivalent to finding a single root of $f(z)$ inside $\mathbb{Y}$.

Since f is monic of degree n over $\mathbb{F}_p$, we can write $f(z) = z^n + c_{n-1}z^{n-1} + \dots + c_0, \forall c_i \in \mathbb{F}_p$. Furthermore, we know that an isomorphism φ is a ring homomorphism and fixes $\mathbb{F}_p$. Hence, for any element $\alpha \in \mathbb{X}$:

$$\varphi(f(\alpha)) = \varphi(\alpha^n + c_{n-1}\alpha^{n-1} + \dots + c_0) = \varphi(\alpha)^n + c_{n-1}\varphi(\alpha)^{n-1} + \dots + c_0 = f(\varphi(\alpha)). \quad (1)$$

Applying this to $\alpha = x$ and using $f(x) = 0$ by definition of $\mathbb{X}$, we get $0 = \varphi(0) = \varphi(f(x)) = f(\varphi(x))$ in $\mathbb{Y}$. **Hence $\varphi(x)$ is a root of f in $\mathbb{Y}$.**

“Conversely, let $y_0 \in \mathbb{Y}$ be any root of f . Every element of $\mathbb{X}$ can be written uniquely as $a(x)$ for a polynomial a of degree $< n$ over $\mathbb{F}_p$; define $\varphi(a(x)) = a(y_0)$ for every such element. This map is well defined (if $f \mid a - b$ then $a(y_0) - b(y_0) = 0$ since $f(y_0) = 0$), it is a ring homomorphism, and it is injective since $\mathbb{X}$ is a field; as $|\mathbb{X}| = |\mathbb{Y}| = p^n$, it is a bijection, hence an isomorphism. Therefore determining an isomorphism is exactly determining one root y_0 of f in $\mathbb{Y}$.”

Existence and number of roots of f in $\mathbb{Y}$ By construction $\mathbb{X} = \mathbb{F}_p[z]/(f)$ is $\mathbb{F}_{p^n}$, a degree- n Galois extension of $\mathbb{F}_p$ with cyclic group generated by the Frobenius $\sigma(u) = u^p$. The generator x satisfies $f(x) = 0$ with f monic irreducible of degree n , so f is its minimal polynomial; since $\deg f = n$, x lies in no proper subfield, hence $f = \prod_{i=0}^{n-1} (z - \sigma^i(x)) = \prod_{i=0}^{n-1} (z - x^{p^i})$. The n conjugates are distinct (otherwise x would lie in a proper subfield), so f has exactly n distinct roots, all in $\mathbb{F}_{p^n}$. As $\mathbb{Y} \cong \mathbb{F}_{p^n}$, f likewise splits completely over $\mathbb{Y}$ into n distinct linear factors, so the algorithm only ever extracts linear factors.

Algorithm Since all of the factors are linear, we can skip the square-free factorisation and distinct-degree factorisation steps and move straight to the equal-degree factorisation step (Cantor-Zassenhaus algorithm), with the factor degree known to be 1.

Algorithm 1 FINDROOT(f) — one root of f in $\mathbb{Y}$

Require: monic $f \in \mathbb{Y}[w]$ splitting into distinct linear factors over $\mathbb{Y}$

Ensure: a root $y_0 \in \mathbb{Y}$ of f

```

1: while deg  $f > 1$  do
2:   repeat
3:      $h \leftarrow \text{CANTORZASSENHAUS}(f)$ 
4:   until  $h \neq \text{FAILURE}$ 
5:    $f \leftarrow h$  ▷ descend into one proper factor
6: end while
7: make  $f$  monic; write  $f = w - y_0$ 
8: return  $y_0$ 
```

This algorithm repeatedly calls the Cantor-Zassenhaus algorithm as described on slide 27 of the lecture notes, without the extra step of calculating the gcd because in our case, we know all roots like in $\mathbb{F}_q$. Each subsequent call to Cantor-Zassenhaus returns a factor of the submitted $f(x)$ or FAILURE with chance $\leq 1/2$. This factor is not guaranteed to be linear yet, so we repeat this procedure until it is.

Complexity Let $M(n)$ be the cost in $\mathbb{F}_p$ -operations of multiplying two degree- $< n$ polynomials over $\mathbb{F}_p$. One multiplication in $\mathbb{Y}$ costs $\mathcal{O}(M(n))$, so one in $\mathbb{Y}[w]/(f)$ costs $\mathcal{O}(M(n)^2)$. Root finding over a field of size Q costs $\mathcal{O}(M(n) \log n \log(nQ))$ operations in that field; here the field is $\mathbb{Y}$ with $Q = p^n$, so $\log(nQ) = \Theta(n \log p)$. Converting each $\mathbb{Y}$ -operation to $\mathbb{F}_p$ -operations (a factor $\mathcal{O}(M(n))$) yields $\mathcal{O}(n \log p \cdot \log n \cdot M(n)^2)$ $\mathbb{F}_p$ -operations, i.e. polynomial in n and $\log p$.

Finding other isomorphisms The cost of finding other isomorphisms is almost free once we have one root y_0 , as the remaining $n - 1$ roots of f in $\mathbb{Y}$ are its Frobenius conjugates $y_0, y_0^p, \dots, y_0^{p^{n-1}}$. This requires only exponentiation.

1.2 Implementation of FieldIsomorphism(f, g)

The implementation can be found as FIELDISOMORPHISM(F, G). The results of running the program with $p = 2^{16} + 7$ and $n = 2^k$ for $k = 2, \dots, 6$ are reported in Table 1.

n	4	8	16	32	64
time (s)	0.000	0.020	0.090	0.860	10.110

Table 1: Running time of `FIELDISOMORPHISM` for $p = 2^{16} + 7$ and $n = 2^k$, $k = 2, \dots, 6$.

1.3 Implementation of `InverseFieldIsomorphism(f, g, s)`

An implementation of `INVERSEFIELDISOMORPHISM` can be found in `GustDeWit_magma.m` as `INVERSEFIELDISOMORPHISM`. This algorithm relies on the fact that φ is not only a field isomorphism, but also an $\mathbb{F}_p$ -linear map between the two $\mathbb{F}_p$ -vector spaces $\mathbb{X}$ and $\mathbb{Y}$ with basis $\{1, x, x^2, \dots, x^{n-1}\}$ and $\{1, y, y^2, \dots, y^{n-1}\}$ respectively.

We know $\varphi = s(y) = y_0$, and since it is a ring homomorphism, $\varphi(x^i) = \varphi(x)^i = y_0^i$, the image of the i -th basis vector of $\mathbb{X}$ is $y_0^i \in \mathbb{Y}$. We can write each y_0^i as its coordinate vector in $\mathbb{Y}$'s basis, resulting in an $n \times n$ matrix M over $\mathbb{F}_p$.

Since φ is a bijection, M is invertible, and M^{-1} is the matrix of the inverse map $\psi = \varphi^{-1}$. Applying M^{-1} to the generator y yields a polynomial t in $\mathbb{X}$'s basis, with $\psi(y) = t(x)$.

2 A Cryptosystem Based on the FFIP

The system as described in the assignment was implemented in `GustDeWit_magma.m` as `SAMPLECHI`, `FFIPKEYGEN`, `FFIPENCRYPT`, and `FFIPDECRYPT`.

3 Attacking the FFIP Cryptosystem (the stupid way)

3.1 Known plaintext-noise-ciphertext-attack

The attack Recall from Section 1.3 that ϕ is an $\mathbb{F}_p$ -linear map $\mathbb{X} \rightarrow \mathbb{Y}$, represented by an invertible $n \times n$ matrix M over $\mathbb{F}_p$ with respect to the bases $\{1, x, \dots, x^{n-1}\}$ and $\{1, y, \dots, y^{n-1}\}$. For every sample write $w_i = m_i + 2r_i = \sum_{j=0}^{n-1} w_{i,j} x^j$ with $w_{i,j} \in \mathbb{F}_p$; then $c_i = M w_i$, where c_i and w_i are coordinate column vectors of length n . Stack the n samples as columns: $C = M * W$, with $W = [w_1 | w_2 | \dots | w_n]$ and $C = [c_1 | c_2 | \dots | c_n]$. If W is invertible over $\mathbb{F}_p$, then $M = C * W^{-1}$.

From M , recover $\phi(x)$ as the image of the basis $x = [0, 1, 0, \dots, 0]^T$, whose image is the coefficient vector $s(z)$.

Probability of failure The probability that this attack fails is the probability that W is singular over $\mathbb{F}_p$. This is dependent on the distribution of the random noise added. The values w_i can be is very limited, with entries $m_{i,j} \in \{0, 1\}$ and $r_{i,j} \in \{-1, 0, 1\}$. This results in only 6 possible values for w_i : $\{-2, -1, 0, 1, 2, 3\}$, all with the chance of occurring and independent of each other, making all entries of W i.i.d. uniform on a 6-element subset of $\mathbb{F}_p$. The chance of this resulting in a singular matrix is the result of Kahn and Komlós (2001). This paper states

$$\Pr(M \text{ singular}) \rightarrow 1 - \prod_{i \geq 1} (1 - p^{-i}) \quad \text{as } n \rightarrow \infty$$

as an asymptotic limit to the non-singular case, valid for our fixed, non-degenerate distribution μ — the uniform distribution on the support $\{-2, -1, 0, 1, 2, 3\} \subseteq \mathbb{F}_p$.

The underdetermined case With $t < n$ triples, the linear system for M is underdetermined. We can restore uniqueness by exploiting that ϕ is also multiplicative, hence, *all* of M can be derived from a single s , leaving only n unknowns. Each triple gives n scalar equations $c_i = \sum_j w_{i,j} s(y)^j$, but these are now nonlinear in the n unknowns.

Recovering $f(z)$ from ϕ

“ Once the isomorphism ϕ is known as the polynomial $s(z)$ (so $\phi(x) = s(y) \in \mathbb{Y}$), recovering $f(z)$ reuses the argument of Section 1.1. There we showed $\phi(f(x)) = f(\phi(x))$ and hence that $\phi(x)$ is a root of f in $\mathbb{Y}$; equivalently $f(s(y)) = 0$. Since ϕ is a bijection (Section 1.3), $s(y)$ generates $\mathbb{Y}$ over $\mathbb{F}_p$ and therefore lies in no proper subfield, so by the same minimal-polynomial reasoning as in Section 1.1 its minimal polynomial over $\mathbb{F}_p$ has degree exactly n and equals $f(z)$.

We compute it by linear algebra over $\mathbb{F}_p$. Reducing modulo $g(z)$, form the coordinate vectors of $1, s(y), s(y)^2, \dots, s(y)^n$ in the basis $\{1, y, \dots, y^{n-1}\}$ of $\mathbb{Y}$. These $n + 1$ vectors lie in an n -dimensional space, so they admit a linear dependency $\sum_{k=0}^n a_k s(y)^k = 0$; it is unique up to scaling because $s(y)$ has a degree- n minimal polynomial. Normalising so that $a_n = 1$ yields $f(z) = \sum_{k=0}^n a_k z^k$. ”

3.2 Implementation of KnownPlaintextNoiseAttack(g, cs, ms, rs)

An implementation of the attack can be found as `KNOWNPLAINTEXTNOISEATTACK(G, CS, MS, RS)`. The results of running the tests can be found in Table 2. The estimated failure rate according to Kahn and Komlós (2001) is also displayed.

k	p	avg. time (s)	succeeded	fail-rate (emp.)	fail-rate (pred.)
1	3	0.012	585/1000	0.4150	0.4399
2	5	0.012	776/1000	0.2240	0.2397
3	11	0.012	898/1000	0.1020	0.0992
4	17	0.012	949/1000	0.0510	0.0623
5	37	0.012	974/1000	0.0260	0.0278
6	67	0.012	976/1000	0.0240	0.0151
7	131	0.012	990/1000	0.0100	0.0077
8	257	0.013	998/1000	0.0020	0.0039

Table 2: Known-plaintext-with-noise attack (Task 3.b) for $n = 32$, $p = \text{NextPrime}(2^k)$, 1000 trials per prime. “Succeeded” counts runs recovering the exact secret $s(z)$; the empirical failure rate is compared against the Kahn-Komlós prediction $1 - \prod_{i \geq 1} (1 - p^{-i})$.

3.3 Gröbner basis attack from a single ciphertext

Setup We are now given only one ciphertext-plaintext pair (c, m) , but not the noise r . Encryption is $c = \phi(m + 2r)$, so the linear attack of the previous section no longer applies. Instead we follow the non-lattice algebraic attack of Doröz et al. (2017, §2.4.2), which avoids working with ϕ directly by using that it is a field isomorphism.

Applying the inverse isomorphism $\psi = \phi^{-1}$ to the ciphertext gives, inside $\mathbb{X} = \mathbb{F}_p[z]/(f(z))$, $\psi(c) = m + 2r$, i.e. $c(z) \equiv m(z) + 2r(z) \pmod{f(z)}$.

Since $\deg(c - m - 2r) < n = \deg f$, this is the polynomial identity $c(z) = Q(z)f(z) + m(z) + 2r(z)$ in $\mathbb{F}_p[z]$ for an auxiliary quotient Q . The unknowns are the coefficients of r (the noise, in $\{-1, 0, 1\}$), of f (which has the special shape $f = z^n + h(z)$ with $\deg h \leq \lfloor n/2 \rfloor$ and coefficients of h in $\{-1, 0, 1\}$), and of Q . Matching coefficients on both sides yields a system of non-linear polynomial equations.

This leads to an underdetermined system of equations, and after eliminating the isomorphism unknowns by a Gröbner basis, we are left with the problem of finding a small point on an n -dimensional variety.

From an n -dimensional to a zero-dimensional variety We resolve this by encoding smallness as equations rather than as a search condition, taking inspiration from Arora and Ge (2011). A value lies in $\{-1, 0, 1\}$ iff it is a root of $t^3 - t = t(t-1)(t+1)$. We therefore add, for every unknown coefficient r_j of the noise and every unknown coefficient h_k of f , the cubic equation $r_j^3 - r_j = 0$, and $h_k^3 - h_k = 0$.

This adds roughly $3n/2$ cubic equations, each confining its unknown to a 3-element set. The common solution set of the divisibility equations together with these cubic equations is therefore *finite*, i.e. a zero-dimensional variety, instead of the n -dimensional variety of Doröz et al. (2017). Using Buchberger’s algorithm, a Gröbner basis then returns the finitely many candidate pairs (f, r) , from which we keep the one whose f has the correct monic shape.

Complexity Once the system is zero-dimensional, the cost of the Gröbner basis computation is governed by the degree of regularity d_{reg} . With $N = \Theta(n)$ unknowns and $m = \Theta(n)$ equations, the standard estimate is

$$\mathcal{O}\left(\left(m \binom{N + d_{\text{reg}} - 1}{d_{\text{reg}}}\right)^\omega\right), \quad (2)$$

with ω the linear-algebra exponent ($\omega \approx 3$ in practice). For a semi-regular system d_{reg} is the index of the first non-positive coefficient of the Hilbert series

$$H(z) = \frac{\prod_i (1 - z^{d_i})}{(1 - z)^N}, \quad (3)$$

where the dominating high-degree generators are the cubic smallness equations ($d_i = 3$). With only a single ciphertext the system is roughly determined rather than overdetermined, so d_{reg} can grow with n . This is why the attack is feasible only for moderate n .

3.4 Implementation of GrobnerBasisAttack(g, c, m, α, β)

An implementation of the attack can be found in `GROBNERBASISATTACK(G, C, M,  $\alpha$ ,  $\beta$ )`. The attack was tested against $p = 2^{16} + 7$, $\alpha = -1$, $\beta = 1$ and $n = 2, 3, 4$ and the results can be found in Table 3.

n	avg. time (s)	avg. #valid keys	valid	equals secret
2	0.001	2.7	10/10	5/10
3	0.005	1.9	10/10	7/10
4	0.356	1.4	10/10	7/10

Table 3: Single-ciphertext Gröbner-basis attack for $p = 2^{16} + 7$, $\alpha = -1$, $\beta = 1$, 10 trials per n .

Comparison against exhaustive search We’re looking for the secret pair (f, s) , however, s is recoverable from f , analogous to Section 1.1. So really we’re only looking for $f = z^n + h(z)$ with degree $h \leq \lfloor n/2 \rfloor$ and each coefficient of h in $\{-1, 0, 1\}$. So the searchspace for f is $3^{\lfloor n/2 \rfloor + 1}$, and for each candidate we must: (i) test irreducibility, (ii) construct the isomorphism s ($\mathcal{O}(n \log p \cdot \log n \cdot M(n)^2)$), (iii) check it’s consistent with the known pair (c, m) by verifying the smallness of r .

Compared to Gröbner, at the tested sizes ($n = 2, 3, 4$) neither is significantly faster. However, their asymptotic behaviour does differ, and at large n , if d_{reg} stays small, Gröbner won’t scale exponentially while exhaustive search will.

Correctness Also visible in Table 3 is that we do not always get back the correct secret key. This is because, by chance, multiple possible solutions fit the constraints layed out by the equations. In practice, this could be minimized by adding a second ciphertext-plaintext pair encrypted with the same key and only accepting a key which worked on both. This practically rules out the existence of other correct solutions.

4 Breaking Search FFIP (the clever way)

4.1 Representation-independence of the trace

For $a \in \mathbb{F}_{p^n}$ the trace is $\text{Tr}(a) = \sum_{k=0}^{n-1} a^{p^k} = \sum_{k=0}^{n-1} \sigma^k(a)$, with $\sigma(u) = u^p$ the Frobenius generator of the Galois group defined in Section 1.1. We show that for any field isomorphism $\phi : \mathbb{X} \rightarrow \mathbb{Y}$ and any $a \in \mathbb{X}$, $\text{Tr}_{\mathbb{Y}}(\phi(a)) = \text{Tr}_{\mathbb{X}}(a)$.

Write $\sigma_{\mathbb{X}}(u) = u^p$ and $\sigma_{\mathbb{Y}}(v) = v^p$ for the Frobenius maps on $\mathbb{X}$ and $\mathbb{Y}$. We now use that ϕ commutes with Frobenius, so $\phi(u^p) = \phi(u)^p$. Using additivity of ϕ ,

$$\phi(\text{Tr}_{\mathbb{X}}(a)) = \phi\left(\sum_{k=0}^{n-1} a^{p^k}\right) = \sum_{k=0}^{n-1} \phi(a^{p^k}) = \sum_{k=0}^{n-1} \phi(a)^{p^k} = \text{Tr}_{\mathbb{Y}}(\phi(a)).$$

Finally, we know $\phi(\text{Tr}_{\mathbb{X}}(a)) = \text{Tr}_{\mathbb{X}}(a)$. Combining with the previous equation, $\text{Tr}_{\mathbb{X}}(a) = \phi(\text{Tr}_{\mathbb{X}}(a)) = \text{Tr}_{\mathbb{Y}}(\phi(a))$. $\square$

4.2 Implementation of DualBasis(basis)

An implementation which computes the dual basis can be found in `DUALBASIS(BASIS)`. Finding the dual basis is simply solving one linear system. Expand each unknown dual vector $\hat{\omega}_j$ in the power basis $\{1, \theta, \dots, \theta^{n-1}\}$ of $\mathbb{F}_{p^n}$ as $\hat{\omega}_j = \sum_{k=0}^{n-1} u_{j,k} \theta^k$. $\text{Tr}(\omega_i \hat{\omega}_j) = \delta_{i,j}$ becomes

$$\sum_{k=0}^{n-1} u_{j,k} \text{Tr}(\omega_i \theta^k) = \delta_{i,j}, \quad \text{i.e.} \quad T U = I_n,$$

with the trace matrix $T_{i,k} = \text{Tr}(\omega_i \theta^k)$ over $\mathbb{F}_p$. Hence $U = T^{-1}$, and column j of T^{-1} , read back in the power basis, gives $\hat{\omega}_j$.

4.3 A polynomial-time attack on search FFIP

The Attack With $C(y) = \sum_{j=0}^{n-1} d_j y^j$ the unknown image of one dual-basis vector, each sample $A(y)$ gives a known linear form $L(d_0, \dots, d_{n-1}) = \text{Tr}(A(y) C(y)) \in [\alpha, \beta]$. As in Section 1.1’s Arora–Ge step, “ $L \in [\alpha, \beta]$ ” is the root condition

$$\prod_{v=\alpha}^{\beta} (L(d_0, \dots, d_{n-1}) - v) = 0, \quad (4)$$

so each sample yields one such equation and N samples give N of them. Solving this system for the d_j recovers $C(y)$; repeating for every dual index recovers all $C_i(y) = \phi(c_i(x))$, hence the dual basis of ϕ , hence ϕ and the secret key.

Degree of the equations Each factor $L - v$ is linear in the d_j , so each equation (4) has total degree $D = \beta - \alpha + 1$. For the project ($\alpha = -1, \beta = 1$) this is the cubic $L^3 - L = 0$, degree 3.

Degree of regularity With n unknowns and N equations all of degree D , the system is strongly overdetermined.

For a semi-regular system d_{reg} is the index of the first non-positive coefficient of $H(z) = \frac{(1-z^D)^N}{(1-z)^n}$, which is the specialised case of equation (3) where all N generators share the single degree $d_i = D$ and the number of unknowns is n . Evaluating this for $n = 32, \alpha = -1, \beta = 1$ ($D = 3$):

N	32	100	1000	10000
d_{reg}	65	11	5	3

At $N = n$ the system is barely determined and $d_{\text{reg}} \approx 2n$ (cost exponential). As N grows d_{reg} drops to its floor D .

Ideal number of samples and complexity Both quantities below rest on one elementary count: the number of monomials of degree d in n variables is $\binom{n+d-1}{d}$, i.e. the number of ways to split d powers among n variables.

Samples. As seen above, more samples lower d_{reg} until it bottoms out at its floor D . This floor is reached once there are at least as many equations as monomials of degree D , i.e. once $N \geq \binom{n+D-1}{D}$. So one would like $N_{\text{ideal}} = \binom{n+\beta-\alpha}{\beta-\alpha+1} = \Theta(n^{\beta-\alpha+1})$, which for $\alpha = -1, \beta = 1$ is $\binom{n+2}{3}$ (e.g. 5984 for $n = 32$).

Complexity. The Gröbner cost is the same bound (2) already established for Task 3.d. The new point here is that N_{ideal} samples force $d_{\text{reg}} = D$, so $\binom{n+d_{\text{reg}}-1}{d_{\text{reg}}} = \Theta(n^D)$ and (2) becomes $\mathcal{O}(n^{D\omega}) = \Theta(n^{3\omega})$. Together with the $\Theta(n^3)$ samples and the $\log p$ factors for $\mathbb{F}_p$ -arithmetic, the whole attack is polynomial in n and $\log p$.

Number of solutions The n dual images $C_1(y), \dots, C_n(y)$ each satisfy (4), but they are not the only solutions. For the project's symmetric interval $[\alpha, \beta] = [-1, 1]$, if $\text{Tr}(AC_i) \in \{-1, 0, 1\}$ then $\text{Tr}(A(-C_i)) \in \{-1, 0, 1\}$ as well, and $C = 0$ trivially satisfies every equation. Hence the solution set is $\{0\} \cup \{\pm C_1, \dots, \pm C_n\}$, i.e. $2n + 1$ solutions.

4.4 Implementation of SearchFFIAttack(samples, α , β)

From the solutions to the key An implementation can be found in SEARCHFFIATTACK(SAMPLES, α , β). The solutions are field elements $C \in \mathbb{Y}$. The secret $f(z)$ is recovered without ever forming ϕ explicitly, using two facts:

1. The solution set $\{\pm C_1, \dots, \pm C_n\}$ is, up to sign, the ϕ -image of the dual basis of $\{1, x, \dots, x^{n-1}\}$, hence is itself the dual basis of $\{\phi(1), \dots, \phi(x^{n-1})\} = \{1, s, \dots, s^{n-1}\}$. So applying DUALBASIS returns $\{1, s, \dots, s^{n-1}\}$.
2. Second, $s = \phi(x)$ is a root of f in $\mathbb{Y}$ and its minimal polynomial over $\mathbb{F}_p$ is $f(z)$ (Section 1.1).

Verifying the key A candidate is accepted only if it is monic irreducible of degree n with the scheme's shape $f = z^n + h(z)$, $\deg h \leq \lfloor n/2 \rfloor$, all coefficients of h in $\{-1, 0, 1\}$. This shape test reliably rejects ill-shaped solutions, so the attack always returns a *working* key. It does not by itself single out the *original* key: several distinct f can satisfy the shape and still fit one sample batch, so the returned key equals the secret only part of the time. Disambiguating the true key requires a second sample batch, or a known (c, m) pair to decrypt and verify the noise is small.

Results Run for $p = 2^{16} + 7, \alpha = -1, \beta = 1$ with the ideal sample count $N = \binom{n+2}{3}$ from Section 4.3, 10 trials per n can be seen in Table 4. The observed solution count is $2n + 1$, the $\{0\} \cup \{\pm C_i\}$ set predicted above.

n	N	avg. time (s)	avg. #solutions	valid	equals secret
4	20	0.001	9.2	10/10	6/10
8	120	0.011	17.0	10/10	8/10
16	816	0.386	33.0	10/10	5/10

Table 4: Search-FFIP attack for $p = 2^{16} + 7, \alpha = -1, \beta = 1$, 10 trials per n .

Effect of $\alpha = 0$ Changing α from -1 to 0 makes the attack *easier*. The equation degree drops to $D = \beta - \alpha + 1 = 2$, so each equation is the quadratic $L(L - 1) = 0$ instead of the cubic $L^3 - L$. The degree of regularity and Gröbner cost shrink accordingly, and the ideal sample count falls from $\binom{n+2}{3}$ to $\binom{n+1}{2}$. The number of solutions also drops: the interval $[0, 1]$ is no longer symmetric about 0, so $-C_i$ is no longer forced to be a solution, leaving essentially the n genuine dual images (plus the trivial $C = 0$).

References

- Arora, Sanjeev, and Rong Ge. 2011. *New Algorithms for Learning in Presence of Errors*, 403–415. January. https://doi.org/10.1007/978-3-642-22006-7_{-}34. https://doi.org/10.1007/978-3-642-22006-7_34.
- Doröz, Yarkin, Jeffrey Hoffstein, Jill Pipher, Joseph H. Silverman, Berk Sunar, William Whyte, and Zhenfei Zhang. 2017. *Fully Homomorphic Encryption from the Finite Field Isomorphism Problem*. Cryptology ePrint Archive, Paper 2017/548. <https://eprint.iacr.org/2017/548>.
- Kahn, Jeff, and János Komlós. 2001. “Singularity Probabilities for Random Matrices Over Finite Fields.” *Combinatorics Probability Computing* 10, no. 2 (March): 137–157. <https://doi.org/10.1017/s096354830100462x>. <https://doi.org/10.1017/s096354830100462x>.